Degree conversion and the finite rank iteration

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Status. This is a bounded author proof awaiting independent review. It proves the degree conversions and the assembly of the rank steps from the precisely stated one-step geometric/frequency interface. It does not prove the analytic sunflower or commutator hypotheses of GTZ Sections 10–11. The complete inverse theorem remains open. Every internal nilcharacter below has exact unit norm and vertical equivariance for all limit elements under the natural ultrapower interpretation. The main manuscript is not modified.

1 The actual degree-to-rank conversion

Let G/Γ be a connected simply connected nilpotent Lie group with lattice and rational degree filtration $(G_i)_{i\geq 0}$ supported on $0 \leq i \leq m$. Apply accepted bounded tautification first, so $G_0 = G_1 = G$ and polynomial orbits may be normalized at zero without changing the represented sequence. Write $\mathfrak{g}_i = \log G_i$. All the vector spaces here are rational for the rational structure determined by Γ.

For $(d, r) \in \mathrm{DR} = \{(d, r) \in \mathbb{N}^2 : 0 \leq r \leq d\}$ with $d > 0$ define $\mathfrak{h}_{(d,r)}$ to be the real span of all iterated Lie brackets of length $\ell \geq 1$ with entries $X_j \in \mathfrak{g}_{i_j}$, $i_j \geq 1$, for which

$$w := \sum_{j=1}^{\ell} i_j > d, \qquad \text{or} \qquad w = d \text{ and } \ell \geq \max(r, 1). \tag{1}$$

Take every bracketing and set $\mathfrak{h}_{(0,0)} = \mathfrak{g}$, $H_{(d,r)} = \exp \mathfrak{h}_{(d,r)}$. Only finitely many nonzero brackets occur, since their weight exceeds m when they can no longer contribute and every entry has positive weight.

Lemma 1. *These are connected rational groups forming a taut degree–rank filtration supported on (m, m). For each $d > 0$, $H_{(d,0)} = H_{(d,1)} = G_d$. Hence the original degree-polynomial orbits are exactly the degree–rank polynomial orbits for this refinement. A degree-m nilcharacter is a degree–rank (m, m) nilcharacter with the same function and with the top character restricted.*

Proof. All bracket spans in (1) are rational, because brackets of rational vectors are rational. Bracketing with $\mathfrak{g}_1$ raises weight by at least one, so each span is an ideal of $\mathfrak{g}$. It is therefore a Lie subalgebra, and its exponential is a connected normal subgroup, which is closed by the global exponential homeomorphism. Choose a rational Mal'cev basis for G/Γ compatible with its rational structure. Each bracket span has a rational basis in these coordinates. Its lower central series is rational, since brackets of rational vectors are rational. Choose one standard Q bounding the finitely many rational structure constants and basis coefficients. Green–Tao, *Annals of Mathematics* 175 (2012), Definition 2.5 and Proposition A.10, applied to this subgroup with its lower central filtration, supplies a Mal'cev basis relative to its actual intersection with Γ. That intersection is

therefore discrete and cocompact. This application uses Lie-algebra rationality in Definition 2.5; it does not assume the lattice conclusion.

The lexicographic nesting is immediate at a fixed d as r increases. If d increases, every new allowed weight was strictly above the old d, so its span is contained in the old one as well. For two positive indices $(d,r),(e,t)$, brackets either have weight greater than $d+e$, or weight exactly $d+e$ and length at least $\max(r,1)+\max(t,1)\geq\max(r+t,1)$. Thus the Lie bracket is in $\mathfrak{h}_{(d+e,r+t)}$. The BCH commutator and its higher terms lie there too, using the ideal property. At index zero use normality. This verifies every pair of indices, not only equal indices.

For $r=0$ or 1, every allowed bracket has weight at least d and thus lies in $\mathfrak{g}_d$. Conversely the length-one brackets with $i_1=d$ include all of $\mathfrak{g}_d$. This proves $H_{(d,0)}=H_{(d,1)}=G_d$. In particular $H_{(1,0)}=G$. If $d>m$ all brackets vanish. At degree m every nonzero bracket has length at most its weight m, which gives the asserted support.

The domain filtration on $\mathbb{Z}$ has $\mathbb{Z}$ at indices $(0,0),(1,0)$ and zero at all other indices, including $(1,1)$, as in Definition 6.17. The derivative criterion for an adapted map therefore asks for its ath differences to lie in $H_{(a,0)}$; these groups equal G_a. This proves the equality of the polynomial classes directly, matching published (6.8). The new top $H_{(m,m)}$ is a rational subgroup of G_m. Restricting the original vertical character is integral on its lattice and preserves all exact vertical identities and unit norm. No enlargement of its character domain is asserted. □

2 The total-degree conversion used in related applications

For a rational multidegree filtration $(G_a)_{a\in\mathbb{N}^q}$ supported on a finite downset J, first use bounded tautification to restrict to $H=\langle G_{e_1},\dots,G_{e_q}\rangle$. Let

$$K_0=H,\qquad \mathfrak{k}_i=\sum_{a\in J:\ |a|\geq i}\mathfrak{g}_a,\qquad K_i=\exp\mathfrak{k}_i\quad(i\geq 1).$$

Each $\mathfrak{g}_a$ is an ideal in $\mathfrak{g}_0$, so the sum is a rational ideal. It equals the Lie algebra of the connected generated group. The brackets satisfy $[\mathfrak{k}_i,\mathfrak{k}_j]\subseteq\mathfrak{k}_{i+j}$, because $[\mathfrak{g}_a,\mathfrak{g}_b]\subseteq\mathfrak{g}_{a+b}$ and $|a+b|=|a|+|b|$. The index-zero condition follows from normality. Also $K_1=H$ and K_i is trivial above $D=\max_{a\in J}|a|$. The rational subgroup lemma verifies every lattice intersection.

For the diagonal restriction $n\mapsto g(n,\dots,n)$ of a multivariable polynomial orbit, use its Taylor expansion. A factor with coefficient $u_a\in G_a$ has scalar exponent $\prod_j\binom{n}{a_j}$ of ordinary degree $|a|$. Such a factor is adapted to $K_\bullet$ since $u_a\in K_{|a|}$; its successive differences have the same one-parameter coefficient and the required scalar polynomial degree. The product is polynomial by B.4. This checks the actual diagonal map rather than treating the domain filtrations as identical. For a rectangle $J=\{a:a\leq d\}$, the only index of total weight $|d|$ is d, so $K_{|d|}=G_d$. A multidegree-d nilcharacter therefore restricts diagonally to a degree-$|d|$ nilcharacter with exactly the same top character. For a nonrectangular downset only the nilsequence degree assertion has been proved, and no unwarranted common top character is claimed.

3 The precise one-step interface

Fix $s\geq 3$, write $m=s-1$, and let $1\leq R\leq m$. Suppose f is bounded on $[N]$, H is a dense internal set of shifts, $A(h,n)$ is a single unit multidegree $(1,m)$ nilcharacter, and (χ_h) is an internal uniformly bounded family of unit degree–rank (m,R) nilcharacters. Suppose that for every retained h, $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A(h,\cdot)\otimes\chi_h$.

The one-step premise used here is the following exact output, after restricting to a dense internal subset and fixing uniform standard bounds. It must be supplied by the corrected additive-quadruple result, the actual Section 10 frequency representation, the two conclusions of Theorem 11.1, accepted LP04 and the final Section 12 join:

1. There is a unit multidegree $(1, m)$ nilcharacter $Z(h, n)$, defined globally, and a family of unit-vector sequences $\zeta_h(n)$ such that $Z(h, n) = \iota\zeta_h(n)$ for retained h, with one fixed coordinate isometry ι. Zero padding is allowed.

2. The entire matrix-valued function $V_h = \chi_h \otimes \overline{\zeta_h}$ is a nilsequence of degree–rank at most $(m, R-1)$, with standard uniform geometry and Lipschitz bounds and internal parameter dependence.

The second clause states a rank drop; for $R > 1$ it does not state an ordinary-degree drop. The final Section 12 constructor must choose its test equivariant on the full ordinary top before invoking LP04's comparison. Equivariance only on $G_{(m,R)}$ does not by itself justify its larger multidegree top. Those obligations remain in the separately authored geometric join and are not hidden in this premise.

4 Recovering the next family without dividing by components

The unit-vector identity gives, for every component a and all relevant integers,

$$(\chi_h)_a(n) = \sum_b (\zeta_h)_b(n)\big((\chi_h)_a(n)\overline{(\zeta_h)_b(n)}\big). \tag{2}$$

The terms in parentheses are components of V_h. Replace the first factor by its corresponding coordinate in $Z(h, n)$ and form the unit nilcharacter $A'(h, n) = A(h, n) \otimes Z(h, n)$ on the product model. The product filtration has the same multidegree bound $(1, m)$.

Choose a scalar component and a degree-at-most-$m-1$ test witnessing the incoming correlation, with fixed bounds and a fixed positive threshold after the usual finite selection. Inserting (2) expresses it as a fixed finite sum of correlations of $\Delta_h f$ against components of $A'(h, \cdot)$, the same ordinary lower-degree test, and a component of V_h. At least one summand has a positive fixed correlation. Finite pigeonhole restricts to a dense subset with fixed component choices if necessary. No scalar component is inverted, or assumed nonzero or of modulus one.

On each fixed rational degree–rank model for V_h, apply the unit-character approximation construction to its top group at $(m, R-1)$. That group is central after tautification, since bracketing with $G_{(1,0)}$ raises the first coordinate above m. Its lattice is rational. Smooth local sections and a finite unit partition on the central torus produce exact-character unit-vector tests. The complex linear span of their components is a unital conjugation-stable algebra separating points; Stone–Weierstrass and a finite uniform net of the bounded Lipschitz test class give a finite approximation by components of such tests. This is the same explicit compact construction in the assembly's selection section, now on the stated degree–rank model. It uses centrality and rationality, not an identification of this top with the larger ordinary-degree top.

Choose the approximation error so that its correlation with the other fixed bounded factors is less than half the surviving threshold. Another finite selection supplies a unit degree–rank $(m, R-1)$ nilcharacter χ'_h for which $\Delta_h f$ has positive degree-at-most-$m-1$ correlation with $A'(h, \cdot)\otimes\chi'_h$. Finite choices and orbit choices are made on representatives and then passed to the ultralimit, retaining internal dependence and uniform bounds. Every test has its exact standard vertical identity before extension, so the all-limit convention is preserved. This is the output required by the next rank step.

5 Initialization, termination and the actual endpoint

Start with the degree-m family obtained from the preceding inverse order. Lemma 1 makes it a degree–rank (m, m) family with the same functions. Take $A = 1$, viewed as a constant unit multidegree $(1, m)$ nilcharacter. The original positive correlation is degree-at-most-$m-1$ correlation with the lower test equal to one.

Apply the one-step premise and the preceding recovery argument at $R = m, m-1, \ldots, 1$. There are exactly m steps, a fixed standard number. A finite sequence of positive standard density losses still leaves a positive standard density; the same holds for positive correlation and finite complexity bounds. Products of the successive multidegree $(1, m)$ nilcharacters retain that bound on their fixed product model.

At the last step the residual family has support at most $(m, 0)$. Its extra degree–rank axiom says $G_{(m,0)} = G_{(m,1)}$. But $(m, 1) > (m, 0)$, so the latter group is trivial. All groups of larger degree are also trivial. Thus the ordinary filtration $(G_{(i,0)})_i$ has degree at most $m-1$. The residual nilcharacter, viewed merely as a bounded nilsequence, can now be multiplied into the already present ordinary degree-at-most-$m-1$ correlation test. The product degree is still at most $m-1$.

We obtain one internal unit multidegree $(1, m) = (1, s-1)$ nilcharacter $A_{\text{final}}(h, n)$ such that $\Delta_h f$ has degree-at-most-$s-2$ correlation with it on a dense set of shifts. This is the exact Theorem 7.1 endpoint needed by the separately authored symmetry proof. It is conditional on the one-step premise being supplied at every input rank and every constructed family. Acceptance of the finite iteration alone does not establish those analytic premises or the full inverse theorem.

6 Source findings and bounded related-use scope

The actual final printed p. 1262 was inspected visually. In Example 6.11 the multidegree-to-degree group is printed with $|a| \le i$, which includes G_0 at every i and does not give bounded degree. For example $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ would give the group $\mathbb{R}$ at every index. The required rule is $|a| \ge i$, proved in Section 2 above.

The same example's degree-to-rank rule prints the equality of summed weights to the fixed bound d, rather than to the running index d'. The desired identity $H_{(d',0)} = G_{d'}$ then need not hold. For instance the linear filtration $G_0 = G_1 = \mathbb{R}$, $G_2 = 0$ with allowed degree bound $d = 2$ yields zero at $(1, 0)$ under that printed rule, whereas $G_1 = \mathbb{R}$. A nonconstant linear polynomial is then lost. Section 1 supplies the current-index rule and proves every filtration, lattice and polynomial assertion used here. The printed length condition is $\ell \ge \max(r', 1)$; that sign is not itself an error.

The bounded related-use scope is Example 6.11, the polynomial identity (6.8), Lemma 6.21's degree conversions, the initialization and rank-zero termination of Theorems 7.1–7.2, and the diagonal/total-degree conversions used in E.9 and the Section 13 assembly. Separate authors of the latter consumers have been notified. No claim is made here to have read every unrelated use of a filtration in the source or every historical correction. These formula errors are not evidence that the intended inverse theorem is false. Earlier source bytes and decisions are preserved.